

Repeated comparison columns force a larger support

Asymptotic research note; independently reviewed

30 September 2026

Let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing, and give the ordered rows arbitrary positive weights. Suppose

$$\mathbb{E} \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

This is the necessary comparison-tensor condition for a distinguished die with K faces in a permutation-fair family of $m + 1$ dice.

Theorem 1. *Suppose at least $r \geq 2$ columns equal a common function a . There is an integer $R \in [r, m]$ such that a has the uniform-interval moments through degree R , and either*

1. $R = m$, so a itself is a scalar quadrature of degree m ; or
2. $R < m$, and

$$K \geq \left\lfloor \frac{R}{2} \right\rfloor (m - R) + \left\lceil \frac{R}{2} \right\rceil + 1.$$

If a is strictly increasing, one can take R to be exactly the number of columns equal to a . The theorem also permits arbitrary plateaus in every column.

Proof. For every column outside the original equal group put $e_j = p_j - a$. The first two mixed moments, using two copies of a , give

$$\mathbb{E} e_j = \mathbb{E} a e_j = 0, \quad \mathbb{E} e_i e_j = 0 \quad (i \neq j).$$

Call e_j flat if $\mathbb{E}(e_j \mid a) = 0$. On every level block B of a , all columns remain nondecreasing, and a flat error has mean zero. For a flat e_i and any other error e_j ,

$$\mathbb{E}(e_i e_j \mid B) = \text{Cov}(e_i, e_j \mid B) \geq 0.$$

Their unconditional inner product is zero. Positivity of the row weights therefore makes this conditional covariance zero on each block. The equality case of the monotone covariance inequality says that at least one function is constant on the block: its covariance is

$$\frac{1}{2} \sum_{s,t \in B} w_s w_t (e_i(s) - e_i(t))(e_j(s) - e_j(t)),$$

and its first-last term is positive if both functions vary.

Consequently, wherever a flat error varies, every other error is constant. Moreover at most one flat error varies on a block; a flat error that is constant on a block is zero there. Replacing all flat columns by a preserves every mixed moment: on each block a product either remains unchanged or has one varying mean-zero error times constant other factors.

Let R be the size of the enlarged equal group, and let $k = m - R$ errors remain. If $k = 0$, the tensor is scalar after replacement, and all moments through degree m follow. Otherwise every remaining error has nonzero conditional mean given a . Choosing distinct columns from the enlarged group gives

$$\mathbb{E}a^s = \frac{1}{s+1} \quad (0 \leq s \leq R), \quad \mathbb{E}a^s e_j = 0 \quad (0 \leq s \leq R-1),$$

and, for distinct remaining errors,

$$\mathbb{E}a^s e_i e_j = 0 \quad (0 \leq s \leq R-2).$$

The signed measure obtained by grouping e_j over the distinct values of a is nonzero and annihilates polynomials of degree at most $R-1$. It must therefore be supported at at least $R+1$ distinct values, by the Vandermonde determinant. In particular a has at least $R+1$ values, and every remaining e_j is nonzero on at least $R+1$ of its level sets.

Put $d = \lfloor (R-2)/2 \rfloor$. The k subspaces

$$U_j = \text{span}\{e_j, ae_j, \dots, a^d e_j\}$$

are mutually orthogonal, each of dimension $d+1 = \lfloor R/2 \rfloor$. Their dimensions follow by evaluating any vanishing polynomial combination on the more than d distinct levels supporting e_j . They are also orthogonal to

$$P = \text{span}\{1, a, \dots, a^{R-1-d}\}, \quad \dim P = R - d = \lceil R/2 \rceil + 1.$$

Indeed all inner products involved have total power at most $R - 2$ for two error spaces and at most $R - 1$ for P and an error space. These orthogonal spaces lie in the K -dimensional row space, giving the claimed bound.

If a is strictly increasing, conditioning on a fixes the row, so a flat error is identically zero. Only columns already equal to a are added, as asserted. $\square$

Corollary 1 (Equal-weight asymptotic consequences). *For equal row weights there is an absolute constant $c > 0$ such that every tensor as above with $r \geq 2$ identical columns satisfies*

$$K \geq cmr.$$

If the entries are rational, the same R from Theorem 1 also satisfies

$$K \geq 2^{2\lfloor R/2 \rfloor}.$$

In particular, a rational comparison tensor with $K \leq m^C$ for a fixed constant C can contain only $O(\log m)$ identical columns. When a repeated class has size of order $\log m$, the new dimension estimate adds an $\Omega(m \log m)$ restriction in the regime $R \leq m/2$.

Proof. Equal-weight interval quadrature of degree R requires $K \geq c_0 R^2$ for an absolute $c_0 > 0$, by the classical Chebyshev quadrature lower bound. If $R \geq m/2$, this implies $K \geq (c_0/2)mR$. If $R < m/2$, the dimension bound gives $K \geq \lfloor R/2 \rfloor(m - R) \geq mR/6$. Since $R \geq r$, the first conclusion follows. The arithmetic conclusion uses the independently proved optimized 2-adic bound in `../independent_directions/optimized_two_adic_filter.tex`. $\square$

The classical real quadrature input is recorded, for example, in Kuijlaars, *The minimal number of nodes in Chebyshev type quadrature formulas* (1993), doi:10.1016/0019-3577(93)90007-L. This note does not prove a universal quadratic per-die lower bound: models with many distinct comparison columns remain outside its effective range. Its conclusion applies to actual dice whenever their comparison columns satisfy the stated repetition hypothesis.